Introduction to Electricity and Electronics

Student Guide

Prepared by BLIS GLOBAL Robotics Department

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Week 2: Understanding Voltage, Current, and Resistance

Introduction to Key Concepts

1. **Voltage (V):**
   - **What is it?**
     - Voltage is the potential difference between two points in an electrical circuit. Think of it as the force that pushes electric charges through a conductor, like the pressure that pushes water through a hose.
   - **Unit of Measurement:**
     - Volts (V)
2. **Current (I):**
   - **What is it?**
     - Current is the flow of electric charge through a conductor. It's similar to the amount of water flowing through a hose per second.
   - **Unit of Measurement:**
     - Amperes (A)
3. **Resistance (R):**
   - **What is it?**
     - Resistance is the opposition to the flow of current in a circuit. Imagine a constriction in a hose that limits the flow of water. That's what resistance does to electric current.
   - **Unit of Measurement:**
     - Ohms (Ω)

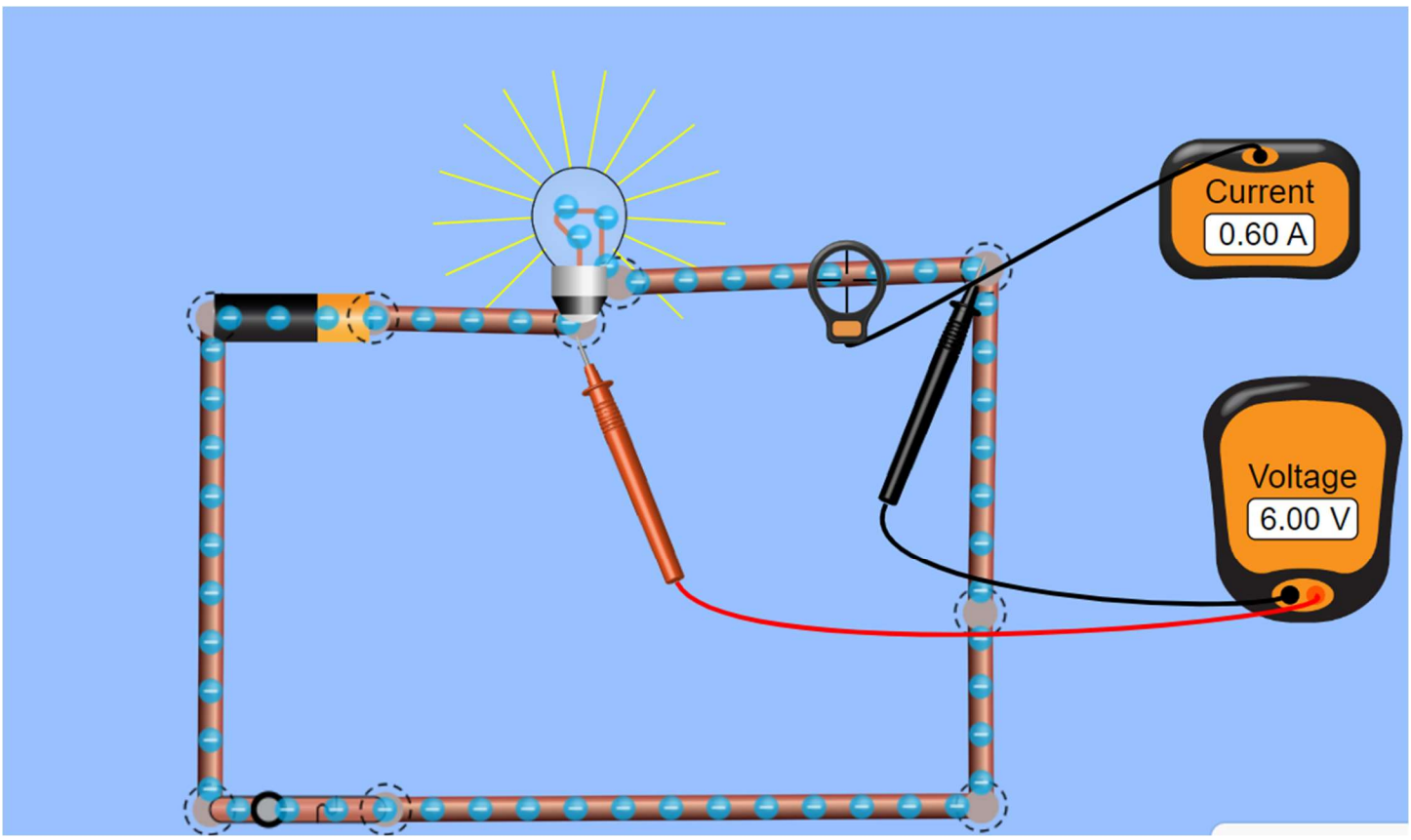
Understanding the Relationship: Ohm's Law

- **Ohm's Law Formula:**
  - V=I×RV = I \times RV=I×R
- **How It Works:**
  - Voltage (V) is the product of Current (I) and Resistance (R). This law shows how these three are related. For example:
    - If you increase the voltage while keeping resistance the same, the current increases.
    - If you increase the resistance while keeping voltage the same, the current decreases.

Practical Demonstration: Seeing Ohm's Law in Action

- **What You'll Need:**
  - A battery, a resistor, an ammeter, a variable power supply, and a multimeter.
- **Steps:**

1. **Set Up the Circuit:**
   - Connect the battery, resistor, and ammeter in a simple series circuit.



Picture 1: Measurement of Voltage and Current

2. **Measure Voltage and Current:**
   - Use the multimeter to measure the voltage across the resistor and the current through the circuit.
3. **Apply Ohm's Law:**
   - Calculate the resistance using the formula R=VIR = \frac{V}{I}R=IV.

4. **Experiment with Voltage:**
    - Increase the voltage and observe how the current changes.
5. **Discussion:**
    - Notice how increasing the voltage causes more current to flow, and how resistance affects this relationship.

Real-Life Applications of Voltage, Current, and Resistance

- **In Your Home:**
    - Devices like your phone charger and TV rely on the right voltage and current to operate safely.
- **In Electronics:**
    - Smartphones, laptops, and other gadgets carefully manage voltage and current to avoid damage.
- **Safety First:**
    - Proper insulation and resistance in wiring protect us from electrical shocks.

Recap & Quick Review

- **What You Learned:**
    - Voltage is like the pressure pushing electricity through a wire.
    - Current is the flow of electricity.
    - Resistance slows down the flow, like a narrow pipe restricts water flow.
- **Quick Questions:**
    - What happens to the current if you increase the resistance but keep the voltage the same?
    - How do voltage and current work together in your everyday electronics?

This lesson gives you a clear understanding of the basic building blocks of electricity: voltage, current, and resistance, and how they interact in an electrical circuit.